

Title: Gerbes of Coassociative Submanifolds and the first Chern class Abstract: We will define (bundle) gerbes, a generalization of principal S^1 -bundles, and define connections on gerbes, whose corresponding forms over a trivialization are 2-forms and whose curvatures are 3-forms. Then, we will build a gerbe from coassociative submanifolds of a G_2 manifold, and study its analogue of the first Chern class, an integral cohomology class in degree 3.

Motivation:

describe motivation in g2 geometry, define G_2 manifolds

ALL MANIFOLDS ARE CONNECTED

Gerbes

(Bundle) gerbes are things like functions $M \rightarrow S^1$ and line bundles over M that live in degree 3, just like the latter two live in degree 1 and 2.

For $f : M \rightarrow S^1$, if we take χ a generator of $H^1(S^1, \mathbb{Z})$, $f^{-1}(c)$ is a codim 1 submanifold of M poicare dual to $f^*\chi \in H^1(M, \mathbb{Z})$, and f is the same thing as gluing together a bunch of functions $g_\alpha : U_\alpha \rightarrow S^1$ over an open cover $\{U_\alpha\}$ satisfying the cocycle condition $g_\beta g_\alpha^{-1}$, so we get a Čech cocycle in $H^0(M, \mathcal{C}^\infty(S^1))$.

Next, we could consider a unitary line bundle $L \rightarrow M$. It has a degree 2 cohomology class $c_1(L) \in H^2(M, \mathbb{Z})$, poicare dual to the zeros of an appropriately chosen section, with transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow S^1$ satisfying $g_{\beta\alpha} = g_{\alpha\beta}^{-1}$ and $g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} = 1$ on $U_\alpha \cap U_\beta \cap U_\gamma$, so we get a Čech cocycle in $H^1(M, \mathcal{C}^\infty(S^1)) \cong H^2(M, \mathbb{Z})$.

A gerbe is this in degree 3:

it's an open covering of M with maps $g_{\alpha\beta\gamma} : U_\alpha \cap U_\beta \cap U_\gamma \rightarrow S^1$ with

$$g_{\alpha\beta\gamma} = g_{\beta\alpha\gamma}^{-1} = g_{\alpha\gamma\beta}^{-1} = g_{\gamma\beta\alpha}^{-1}$$

and

$$dg = g_{\beta\gamma\delta} g_{\alpha\gamma\delta}^{-1} g_{\alpha\beta\delta} g_{\alpha\beta\gamma}^{-1} = 1$$

Thus we get a class in $H^2(M, \mathcal{C}^\infty(S^1)) \cong H^3(M, \mathbb{Z})$ (the proof is via the exponential sheaf sequence).

Remark

From the definition, we can tensor gerbes (take products of cocycles) and pull them back.

How do we trivialize gerbes?

A line bundle is trivialized by a unitary section $s/|s|$ (for $s \in \Gamma(L)$ nonvanishing). By thinking about the associated principal S^1 -bundle, this is a collection of circle valued functions $f_\alpha : U_\alpha \rightarrow S^1$ with $f_\alpha = g_{\alpha\beta} f_\beta$ on $U_\alpha \cap U_\beta$, and so is a degree 1 thing, and then $f'_\alpha / f_\alpha = f'_\beta / f_\beta$, so two trivializations differ by a global function $f : M \rightarrow S^1$.

To trivialize a gerbe, we go a degree up:

A trivialization of a gerbe is defined by functions $f_{\alpha\beta} = f_{\beta\alpha}^{-1} : U_\alpha \cap U_\beta \rightarrow S^1$, such that $g_{\alpha\beta\gamma} = f_{\alpha\beta} f_{\beta\gamma} f_{\gamma\alpha}$, i.e. a representation of a Čech cocycle as a coboundary. On overlaps, letting $h_{\alpha\beta} = f'_{\alpha\beta} / f_{\alpha\beta}$ (so $h_{\alpha\beta}^{-1} = h_{\beta\alpha}$), we have $h_{\alpha\beta} h_{\beta\gamma} h_{\gamma\alpha} = 1$, so the difference of two trivializations is the data to form a line bundle.

In particular, over one open set U_α in our trivializing cover, let $g_{\alpha\beta\gamma} = f_{\beta\gamma}$ so that the cocycle condition gives

$$g_{\beta\gamma\delta} = f_{\beta\gamma} f_{\gamma\delta} f_{\delta\beta}$$

and $f_{\alpha\beta} = 1$, we have a trivialization of a line bundle, which on $U_\alpha \cap U_\beta$ the two trivializations differ by a line bundle $L_{\alpha\beta}$.

use line bundles on $U_\alpha \cap U_\beta$ rather than functions. A gerbe is defined using this language by:

- a line bundle $L_{\alpha\beta}$ on each $U_\alpha \cap U_\beta$
- an isomorphism $L_{\alpha\beta} \cong L_{\beta\alpha}^{-1}$
- a trivialization $\theta_{\alpha\beta\gamma}$ of $L_{\alpha\beta}L_{\beta\gamma}L_{\gamma\alpha}$ (tensor product) on $U_\alpha \cap U_\beta \cap U_\gamma$
- the trivialization $\theta_{\alpha\beta\gamma}$ satisfies $\delta\theta = 1$ on $U_\alpha \cap U_\beta \cap U_\gamma \cap U_\delta$

With regard to the last condition, the point to notice is that $\delta\theta$ is a section of a tensor product of 12 line bundles corresponding to the 12 ordered pairs of elements from $\{\alpha, \beta, \gamma, \delta\}$. Using the isomorphisms $L_{\alpha\beta} \cong L_{\beta\alpha}^{-1}$, the products cancel in pairs and so this becomes a section of the trivial bundle where the trivial section 1 is well-defined.

How does this depend on the choices we made? The problem is that the total space of gerbes aren't manifolds, so the usual global methods (sections) don't really work. Instead, one can interpret these in terms of sheaves of so-called (flat) Dixmier-Douady groupoids, as done by Brylinski, where he essentially uses a homological Chern-Weil theory over such groupoids to show nothing depends on these choices of trivializations (all the categories obtained are equivalent). Thus, for now we will just worry about what we can use gerbes for.

Differential Geometry of Gerbes

A connection on a line bundle is given by a bunch of connection 1-forms A_α defined on each U_α such that $iA_\alpha - iA_\beta = g_{\alpha\beta}^{-1}dg_{\alpha\beta}$, where since the right hand side is closed, there is a global closed 2-form, the curvature F , so that $F|_{U_\alpha} = dA_\alpha$.

Conversely,

Proposition

If M admits a finite good cover (e.g. M is compact) and a closed 2-form F with $[F]/2\pi$ integral, there is a line bundle with connection whose curvature is F .

Proof:

By the Poincaré lemma, there are 1-forms A_α on each U_α with $F|_{U_\alpha} = dA_\alpha$, so that $A_\beta - A_\alpha = 0$ on $U_\alpha \cap U_\beta$, so by the Poincaré lemma again we have functions $f_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow \mathbb{R}$ with $A_\beta - A_\alpha = df_{\alpha\beta}$. If $[F]/2\pi$ is integral (i.e. the image of an integral class under the de Rham homomorphism), we can choose such $f_{\alpha\beta}$ such that $g_{\alpha\beta} := \exp(if_{\alpha\beta})$ is a cocycle, so that we get a line bundle, and then F is the curvature of a connection built from the 1-forms A_α . ■

We can do the same thing in degree 3, to get a closed curvature 3-form G representing an integral class defined by

$$\begin{aligned} G|_{U_\alpha} &= dF_\alpha \\ F_\beta - F_\alpha &= dA_{\alpha\beta} \\ iA_{\alpha\beta} + iA_{\beta\gamma} + iA_{\gamma\alpha} &= g_{\alpha\beta\gamma}^{-1}dg_{\alpha\beta\gamma} \end{aligned}$$

The $A_{\alpha\beta}$'s are connection 1-forms for trivializing line bundles of our gerbe, so this data is the same thing as:

A connection $\nabla_{\alpha\beta}$ on each $L_{\alpha\beta}$ such that over each trivialization $\theta_{\alpha\beta\gamma}$ of $L_{\alpha\beta} \otimes L_{\beta\gamma} \otimes L_{\gamma\alpha}$, $\nabla_{\alpha\beta\gamma}\theta_{\alpha\beta\gamma} = 0$, such that on each U_α , we have a 2-form F_α with $F_{\alpha\beta} = F_\beta - F_\alpha$ on overlaps is the curvature of $\nabla_{\alpha\beta}$.

G₂ Geometry

Let (M, φ, ψ) be a compact connected G₂-manifold.

Definition

A monopole gerbe over M is a gerbe $(\mathcal{G}, F)$ with connection such that

$$\star(F_\alpha \wedge \psi) = d\phi_\alpha$$

for F_α the connection 2-forms and ϕ_α some real-valued functions. (1,0) condition
The curvature of F is the harmonic representative of $c_1(\mathcal{G})$. (HYM condition)
Monopole gerbes form a group under tensor products.

Proposition

Let N be a codimension 3 submanifold of M . Then given a gerbe $\mathcal{G}$ defined by $H = PD(N) \in H^3(M, \mathbb{Z})$, there is a connection F whose curvature is H .

Sketch:

Let δ_N be the current generated by N , so that $[H - \delta_N] = 0$ in de Rham cohomology for currents. Thus because the Hodge decomposition for currents is $\mathcal{D}'(\Omega^k) = \mathcal{H}^k \oplus \text{im } \Delta$, we may distributionally solve $\Delta H_0 = H - \delta_N$. Shrinking things locally if necessary, we can also solve $\Delta H_\alpha = H$. Define connection 2-forms by $F_0 = d^* H_0, F_\alpha = d^* H_\alpha$ (since everything we solved is unique up to harmonic 1-forms, these are well-defined), and since $G = dd^* H_0 = dd^* H_\alpha$, it follows that the curvature of the connection defined this way is H . ■

Theorem

If N is coassociative, i.e. $\varphi|_N \equiv 0$, $(\mathcal{G}, F)$ is a monopole.

Sketch:

Step 1: Observe we defined the curvature of F above to be the harmonic representative of H .

Step 2: Use standard G_2 representation theory and its consequences (i.e. decompositions of forms into relevant types) combined with $\delta_N \wedge \varphi = 0$ (coassociative) to show that the monopole equation $\star(F_i \wedge \psi) = d\phi_i$. For $i = 0$, we have $\phi_0 = 7a$, where $\pi_1(H_0) = -a\varphi$.

Step 3: For $i = \alpha$, $\phi_\alpha = -d^* b_\alpha$, where b_α is the solution to

$$\nabla^* \nabla b_\alpha = 3d^* d^7 b_\alpha + dd^* b_\alpha = -\star(F'_\alpha \wedge \psi)$$

using that the metric induced by φ is Ricci-flat. The fact we can solve this PDE is a minimization argument for the functional $J(u) = \int_U |\nabla u|^2 + \langle u, \star(F_\alpha \wedge \psi) \rangle$. ■

Theorem

The map from monopole gerbes to $H^3(M, \mathbb{Z})$ given by taking curvature of its connection is surjective.

Sketch:

We start with a gerbe $\mathcal{G}$ representing the right class $H^3(M, \mathbb{Z})$, and build the required connection.

Let H be a harmonic representative of $c_1(\mathcal{G})$, and do Step 3 of the argument as above to build a connection via $dF'_\alpha = H$, and define $F_\alpha = F'_\alpha + d(\star(db_\alpha \wedge \psi))$ with b_α solving the same PDE as above. ■

MORE TO SAY IF OUT OF TIME:

Using these results, Oliveira proved topological constraints on G_2 manifolds,

Theorem 3. *Let (X, φ) be a compact G_2 -manifold, $\alpha \in H^3(X, \mathbb{Z})$, and N a coassociative representative of α . If τ and χ respectively denote the signature and Euler characteristic of N , then $p_1(X) \cup \alpha = 6\tau - 2\chi$. In particular,*

- $\tau > \frac{1}{3}\chi$ if and only if $p_1(X) \cup \alpha > 0$;
- $\tau = \frac{1}{3}\chi$ if and only if $p_1(X) \cup \alpha = 0$;
- $\tau < \frac{1}{3}\chi$ if and only if $p_1(X) \cup \alpha < 0$.

Corollary 1. *Any coassociative T^4 of a compact G_2 -manifold must represent a class α such that $p_1(X) \cup \alpha = 0$. In particular, if $b^3(X) = 1$, then (X, φ) has no coassociative tori.*

We must however remark that up to the author's knowledge there are no known examples of G_2 -manifolds with $b^3(X) = 1$.

Moreover, we can relate this construction to the loop space as follows: essentially, using a loop (a map $f : S^1 \rightarrow M$) we can pullback our gerbe, which since S^1 is 1-dimensional, is flat and has trivial holonomy. This gives rise to a certain space acted on freely and transitively by $S^1 \cong H^1(S^1, \mathbb{R}/\mathbb{Z})$. Thus we have a principal S^1 -bundle over the space of loops. The pulled back gerbe then induces a notion of parallel transport, hence a connection on the loop space, and transgresses the connection 3-form to a connection 2-form on this line bundle. So we get normal line bundle things, but it's because we passed a degree down by considering maps $S^1 \rightarrow M$.